

Question:
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Key relationships and values :

- De Broglie wavelength : $\lambda = h/p = h/(\sqrt{2 m E_k})$
- Mass of electron : $m \approx 9.109 \times 10^{-31}$ kg
- Planck constant : $h \approx 6.626 \times 10^{-34}$ J $\cdot$ s = 4.136×10^{-15} eV $\cdot$ s (for consistent units)
- Energy in SI : $E_k = 1$ eV = 1.602×10^{-19} J = 1.602×10^{-12} eV
- Convert mass to cgs : $m \approx 9.109 \times 10^{-30}$ g
- Electron speed is small compared to c , so kinetic approximation $h / \sqrt{2 m E_k}$ is valid .

Compute momentum :

$$p \approx \sqrt{2 m E_k}$$

Energy $\rightarrow$ momentum relation for non relativistic electrons gives $m v = p$, so wavelength (in cm) is
 $\lambda \approx (hc / (E_k e)) \approx (4.136 \times 10^{-15} \text{ eV}\cdot\text{s} \times 2.418 \times 10^9 \text{ cm}\cdot\text{s}^{-1}) / (1.602 \times 10^{-12} \text{ eV}) \approx 6.582 \times 10^{-7} \text{ cm} \approx 6.6 \times 10^{-7} \text{ cm} \approx 6.625 \times 10^{-27} \text{ erg}\cdot\text{s}$

Thus , the de Broglie wavelength of an electron with 1 eV kinetic energy is about 6 .

